Thm. Let f be an analytic on the closed disk $|z-z_0|\le r$. Then
$$f(z_0)=\frac{1}{2\pi}\cdot\int_0^{2\pi} f(z_0+r\cdot e^{i\theta})\,d\theta.$$

Proof. By the Cauchy's integral formula,
$$f(z_0)=\frac{1}{2\pi i}\cdot\int_{|z-z_0|=r}\frac{f(z)}{z-z_0}\,dz$$
Then, for a parameterization $z(\theta)=z_0+r\cdot e^{i\theta}$ $(0\le\theta\le 2\pi)$,
$$=\frac{1}{2\pi i}\cdot\int_0^{2\pi}\frac{f(z_0+re^{i\theta})}{re^{i\theta}}\cdot r\cdot i e^{i\theta}\,d\theta$$
$$=\int_0^{2\pi} f(z_0+re^{i\theta})\,d\theta \qquad \blacksquare$$

This theorem is called the Gauss's Mean-Value Theorem.

Maximum Modulus Theorem.

If non-constant f is analytic in a domain D, then $|f|$ cannot attain a maximum in D.

Proof. Suppose that $|f|$ attains maximum at $z_0\in D$. (Then f must be a constant).
Since D is open, for some $r>0$, $\overline{B_r(z_0)}\subseteq D$. Then, the Gauss mean-value Thm gives
$$|f(z_0)|\le\frac{1}{2\pi}\cdot\int_0^{2\pi}|f(z_0+re^{i\theta})|\,d\theta\le\frac{1}{2\pi}\cdot\int_0^{2\pi}|f(z_0)|\,d\theta=|f(z_0)|$$
$|f(z_0)|$ is maximum.

So that
$$|f(z_0)|=\frac{1}{2\pi}\cdot\int_0^{2\pi}|f(z_0+re^{i\theta})|\,d\theta,\quad\text{or}$$
$$\frac{1}{2\pi}\cdot\int_0^{2\pi}|f(z_0)|-|f(z_0+re^{i\theta})|\,d\theta=0.$$

By the assumption, the term in the integral is non-negative, thus
$$|f(z_0+re^{i\theta})|=|f(z_0)|\quad\text{for all } 0\le\theta\le 2\pi.$$

for all $0<\varepsilon\le r$, $|f(z_0)|=|f(z_0+\varepsilon\cdot e^{i\theta})|$ and so $|f(z)|=|f(z_0)|$ for all $z\in\overline{B_r(z_0)}$.
So, f is constant, the identity theorem gives the result.